API Gateway

→ First of all, serverless!

Simple:

- You are building an app,
- Users & other services talks at ~~emp~~ endpoints called api endpoints.

- These endpoints are managed by api gateway → very very useful.

→ You are building an app where

User send request,
your backend [servers, functions, database] process them

- But we don't want user to directly access the backend, why?

- Security risk,
- Harder to manage 1000s of user,
- No control over who calls what.
- No authentication.

API gatway works as a middleman & solves all these

API gateway does 5 fundamental things

1.] Routing

GET /user → Get the user data from backend

2.] Authentication - Who are you?
3.] Rate limit before backend crashes
4.] Can change input file too
like user submit Text → Backend require JSON
Done by API gateway.
5.] Monitoring & Logging

- Who called
- Which endpoint
- For How long?

Then Mon! what's the difference betn API gateway & Load balancing?

API gateway	Load Balancing
- Works at higher level.	- Works at low level
- Understands route like /users, /orders	- Just distribute the load
- GET, POST	- Health check
- Authentication	- failover
- caching	

Manage & Route 1000s of user

Lamda + API Gateway

⇒ Deadly Million Dollar Combo

⇒ Truely serverless

Performs task & results output on serverless

Types of API in Gateway

HTTPS API	Rest API	WEBSOCKET API
Low cost & lower latency	- High cost, but feature rich	- Real time, two-way communication
- Best for Lamda routing	- Features like, caching, API keys, WAF integration.	
- Billed per request	- Billed per request	- Billed on message & connection time.
$1 per million request	$ 3.50 per million request	